

Student Handout — Common Fault Flowcharts (Text Form)

Use with the 7-step process. These are starting guides — not excuses to skip measurements.

A. “No Go” / Won’t Move (Electric)

1. Verify symptom (key on, direction selected, pedal applied, brake released).
 2. Safety/visual: connectors, burned cables, loose lugs, fault LED.
 3. Measure **pack voltage** at batteries and at solenoid/controller input.
 - Low/unstable → battery/connection diagnosis (Week 2 methods).
 4. Listen/test **solenoid**: coil voltage when commanded; contact drop under attempt-to-run.
 5. Check **controller codes** (Curtis/etc.); address code + confirm with pin tests.
 6. Test **throttle signal** at controller (0→100%).
 7. Verify direction switch / F&R contactors / HPD sequence.
 8. Motor tests only after power/control proven.
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B. “Slow / Weak”

1. Verify on flat ground, correct tire pressure, not in turtle mode if equipped.
 2. Pack OCV + load capability / weak battery hunt.
 3. Voltage drop on main cables & solenoid contacts under load.
 4. Controller current-limit / throttle calibration / fault cutback.
 5. Motor brush/commutator / binding / brake drag.
 6. Mechanical drag (brakes partially applied, underinflated tires, binding bearings).
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C. Charger Issues

1. Confirm outlet AC and cord integrity.
 2. DC connection / polarity / pack presence.
 3. Interlock/enable circuits.
 4. Measure DC output / current.
 5. Chemistry match (LA vs lithium) and BMS enable.
 6. Charger internal fault (escalate; don’t open live).
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D. Intermittent Power

1. History: bump, wet weather, heat, only when turning?

2. Wiggle test connectors **with care** while monitoring voltage.
 3. Inspect chafe points, grounds, pin tension.
 4. Thermal: retest hot.
 5. Do not shotgun controllers first.
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E. Gas No-Start (quick)

1. Safety interlocks / kill switch.
2. Fuel present → filter → delivery.
3. Spark + plug condition.
4. Compression.
5. Air filter / flood condition / OEM procedure.

[DIAGRAM PLACEHOLDER: visual flowchart versions of A-D for shop wall]